SMART SPHERE CLASSROOM

A PROJECT REPORT

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BONAFIDE CERTIFICATE

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ABSTRACT

Smart Sphere Classroom is a next-generation online learning management system designed to address the limitations of traditional educational platforms like Google Classroom, Moodle, and Blackboard. It offers a unified, intelligent, and engaging environment for both students and faculties, enhancing productivity, collaboration, and academic outcomes.

The platform includes an AI chatbot, auto test evaluation, custom notifications, calendar tools, GPA/CGPA calculators, assignment tracking, and classroom folders. The Gen Flash AI chatbot offers real-time doubt support, while instant grading reduces faculty workload. Custom alerts keep students informed. Together, these features streamline academic tasks and enhance learning.

Custom notification alerts, integrated using Email.js, help students stay informed about upcoming deadlines and tasks through a flexible scheduling system. Additionally, gamified learning experiences like quiz games, course learning platforms with certification, and academic calculators have been incorporated to foster interactive learning and self-assessment.

The system is developed using modern technologies such as HTML, CSS, JavaScript for frontend, Node.js with Express.js for backend APIs, MongoDB for database management, and Express JS server for AI chatbot functionalities.
Through seamless integration of cloud technologies and AI automation, Smart Sphere Classroom delivers a comprehensive digital learning experience.

Keywords: AI-powered Chatbot, Custom Notifications, Automatic Test Evaluation, GPA and CGPA Calculator, Assignment Management, Digital Learning.

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CHAPTER 1

INTRODUCTION

1.1 GENERAL INTRODUCTION

The rapid evolution of education, accelerated by global events and digital access, has made online learning a core part of modern education. Learning Management Systems (LMS) like Moodle, Blackboard, and Google Classroom have supported this shift but still face key limitations.

Smart Sphere Classroom is designed to overcome these challenges by providing an intelligent, integrated platform that enhances engagement, simplifies educator tasks, and personalizes learning. Inspired by modern LMS tools, it offers a cohesive solution to avoid fragmentation and inefficiency.

Key Limitations Addressed:

1. Low Personalization: Existing systems lack flexibility to customize learning paths for individual students.
2. Manual Assessment: Educators struggle with time-consuming, non-automated evaluation processes.
3. No Real-Time Support: Students often miss timely help due to the absence of AI-based assistance.
4. Fragmented Communication: Multiple tools cause confusion and reduce collaboration.

1.2 OBJECTIVE OF THE PROJECT

The primary objective of Smart Sphere Classroom is to create an advanced and user-friendly online educational platform that enhances the teaching and learning experience through innovative features.

The specific objectives include:

1) **Streamlined Classroom Management: To** provide functionalities similar to Google Classroom for managing assignments, announcements, and student collaboration within a single, intuitive interface.

2) **Custom Scheduling with Notifications:** To enable educators to set custom reminders and alert notifications at specified intervals, helping students complete tasks on time and stay organized.

3) **Automatic Test Evaluation:** To automate the grading process for quizzes and tests, providing instant feedback and performance analytics to students and educators, saving time and improving the learning process.

4) **AI-Powered Chatbot Support:** To incorporate an intelligent mini-chatbot to assist students with subject-related queries, offering instant clarification and academic support, and reducing the workload on educators.

5) **Improved Productivity and Engagement:** To foster a more efficient, engaging, and collaborative digital classroom environment for both educators and learners.

6) **Seamless Integration:** To combine all features into a single platform, reducing the need for multiple tools and ensuring a cohesive user experience.

1.3 MOTIVATION OF THE PROJECT

Traditional LMS platforms like Moodle, Blackboard, and Google Classroom offer basic features but lack intelligent automation, personalized support, and seamless communication. Smart Sphere Classroom addresses these gaps by integrating:

AI-powered chatbot for instant doubt resolution and personalized guidance.

Automated test evaluation to reduce faculty workload and provide instant feedback.

Custom learning paths with smart notifications and calendar integration.

Gamified learning through quizzes and GPA calculators to make education engaging and interactive.

1.4 PROBLEM STATEMENT

Traditional LMS platforms like Moodle and Google Classroom handle basic tasks but lack intelligent features such as real-time support, automated grading, and effective task management. This results in higher faculty workload, unresolved student doubts, missed deadlines, and low engagement. Smart Sphere Classroom addresses these issues with AI assistance, automation, personalized notifications, and interactive learning tools to boost efficiency and student outcomes.

1.5 SCOPE OF THE PROJECT

The Smart Sphere Classroom project focuses on the development and deployment of an integrated educational platform featuring secure user authentication, assignment management, online test and quiz modules with automated evaluation, and a real-time AI-powered chatbot for academic support. The platform includes custom notification scheduling via Email.js, a classroom folder system for note sharing, and interactive tools like quiz games and CGPA calculators. Built using HTML, CSS, JavaScript, Express.js, MongoDB, and Gen Flash AI 2.0, the system is designed to streamline academic activities and enhance the online learning experience for both students and faculty.

CHAPTER 2

LITERATURE SURVEY

2.1 INTRODUCTION

The development of Smart Sphere Classroom is informed by a wide range of research in the fields of learning management systems, artificial intelligence in education, automated assessment, and personalized learning. This literature review provides an overview of relevant research, highlighting key findings and identifying gaps that Smart Sphere Classroom aims to address.

2.2 LITERATURE SURVEY

SURVEY – 1

TITLE : AI-based Chatbot and NLP

AUTHOR : S. Kumar & R. Patel

YEAR : 2020

ABSTRACT : This project presents an AI-powered chatbot that uses Natural Language Processing (NLP) to offer instant, personalized academic support. It streamlines the query-resolution process, enabling students to access critical information anytime while promoting self-paced learning. The chatbot's accurate interpretation of student queries enhances the overall learning experience and reduces faculty workload. With an intuitive interface and quick response times, it makes academic support more efficient and accessible. Its scalability and flexibility make it a valuable tool for institutions aiming to strengthen their student support systems and modernize digital learning environments.

SURVEY – 2

TITLE : Cloud-based LMS with Automated Assessment

AUTHOR : J. Williams et al.

YEAR : 2021

ABSTRACT : This project presents a cloud-based Learning Management System (LMS) with automated grading and feedback for online tests. It reduces instructor workload while providing instant feedback to students, ensuring timely and accurate evaluations. The system offers scalability, reliability, and accessibility, making online learning more efficient and engaging.

SURVEY – 3

TITLE : Custom Scheduling and Reminder System

AUTHOR : M. Lee & T. Johnson

YEAR : 2019

ABSTRACT : This project introduces a custom scheduling and reminder system to help students stay organized and manage their time effectively. With personalized notifications, students can prioritize tasks, set reminders, and stay updated on deadlines. The system also allows instructors to send reminders and announcements, improving communication and collaboration. By promoting better time management, it enhances academic performance and the overall learning experience.

SURVEY – 4

TITLE : ML-powered Adaptive Learning System

AUTHOR : A. Fernandez et al.

YEAR : 2022

ABSTRACT : This project introduces an adaptive learning system powered by machine learning (ML) algorithms to deliver personalized learning paths based on

student performance, learning style, and preferences. Using ML-driven analytics, it identifies knowledge gaps, evaluates strengths and weaknesses, and adjusts content accordingly. Students benefit from tailored recommendations and feedback, allowing them to learn at their own pace. Instructors can monitor progress, pinpoint areas needing support, and adapt their teaching strategies. This ML-powered system aims to revolutionize learning by making it more effective, efficient, and engaging.

SURVEY – 5

TITLE : Gamification in Online Education

AUTHOR : P. Gupta & K. Sharma

YEAR : 2023

ABSTRACT : This project uses gamification in online education to boost student engagement and learning outcomes. By integrating quizzes, challenges, rewards, and leaderboards, it creates an interactive environment that encourages active participation. Instructors can design gamified content, and students benefit from a more enjoyable and motivating learning experience.

2.3 CONCLUSION

This literature review highlights the importance of integrating advanced technologies like AI, automated assessment, and personalized learning into learning management systems. Research indicates these innovations enhance student engagement and learning outcomes. However, gaps remain, especially in the need for more adaptive systems. Smart Sphere Classroom addresses these gaps by providing a comprehensive learning environment driven by AI, automation, and personalization.

CHAPTER 3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

Learning Management Systems (LMS) such as Moodle, Blackboard, and Google Classroom play a pivotal role in delivering, managing, and assessing educational content. These platforms centralize resource access, facilitate instructor-student communication, and support assessment and feedback mechanisms**.**

MOODLE

Moodle is an open-source LMS designed to help educators create customizable learning environments. Its modular architecture allows institutions to adapt features to specific needs, including course creation, assignment submission, quiz management, and grade tracking. However, advanced features like real-time notifications, AI-powered chatbots, or predictive analytics depend on third-party plugins, which may lead to compatibility issues and require ongoing maintenance (Dougiamas & Taylor, 2003).

BLACKBOARD

Blackboard is a commercial LMS known for its user-friendly interface and built-in communication tools. It supports assignment management, discussions, grading, and feedback with a polished user experience. However, it lacks native AI-driven features such as personalized learning, smart recommendations, or automated scheduling. Institutions often rely on external integrations to add these functionalities (Zanjani et al., 2017).

GOOGLE CLASSROOM

Google Classroom is a lightweight, cloud-based LMS integrated with Google Workspace tools like Docs, Drive, and Calendar. It enables teachers to create and distribute assignments, share resources, and manage class communications efficiently. While ideal for smaller classrooms, it has limited built-in automation for grading and lacks advanced features like predictive analytics or intelligent reminders, making it less suited for large-scale institutional use (Alim et al., 2019).

3.2 DISADVANTAGES OF THE EXISTING

MOODLE:

Dependency on Plugins: Many advanced features like chatbots, real-time alerts, or predictive analytics are not built-in and require third-party plugins, increasing complexity.

Technical Expertise Required: Proper setup, customization, and maintenance often need technical knowledge, making it less user-friendly for non-technical institutions.

Interface Outdated: Without customization, the default user interface can feel outdated and unintuitive compared to modern platforms.

BLACKBOARD:

Lack of Built-in AI Features: Despite being polished, Blackboard does not include smart automation like AI-based tutoring, personalized learning suggestions, or automatic scheduling alerts.

High Cost: As a commercial platform, Blackboard licensing fees are high, making it less accessible for smaller institutions or developing regions.

Complex Navigation: Some users report that the interface can feel overwhelming and confusing, especially for first-time users.

GOOGLE CLASSROOM:

Limited Assessment Features: It lacks automatic evaluation for complex assessments, advanced grading rubrics, or detailed performance analytics.

Basic Functionality: While it covers basic classroom management, it does not support features like competency tracking, course prerequisites, or gamification.

Reliance on Google Ecosystem: Heavy dependency on Google tools means limited flexibility if institutions use other systems.

3.3 PROPOSED SOLUTIONS

The proposed system, **Smart Sphere Classroom**, is an innovative, comprehensive, and user-friendly online educational platform that bridges the gaps identified in traditional Learning Management Systems (LMS).

It aims to integrate classroom management with advanced technologies like **Artificial Intelligence**, **Real-Time Notifications**, **Gamified Learning**, and **Automated Evaluations** to create a holistic digital learning environment.

Smart Sphere Classroom offers a **centralized portal** where:

Faculties can assign, monitor, and evaluate students' academic performance effortlessly. Students can engage in interactive learning activities, clarify doubts instantly, and manage their coursework efficiently. The system architecture is built on **modern web technologies** ensuring **fast, scalable, and responsive operations**.

3.3.1 THE KEY COMPONENTS OF THE PROPOSED SOLUTION INCLUDE:

- **AI-Powered Chatbot:**

A real-time academic assistant that resolves students' doubts instantly, built using the Gen Flash AI model and integrated with backend server.

- **Custom Notification Scheduling:**

Personalized reminders and deadline alerts powered by Email.js to improve time management among students.

- **Assignment Management Platform:**

Faculty can assign work with descriptions and deadlines; students can upload their assignments securely through the portal.

- **Online Test and Quiz Platform with Automated Evaluation:**

Faculties can create quizzes/tests; students can attempt them online; the system evaluates scores automatically and publishes results instantly.

- **Classroom Folder System:**

Faculties can upload learning resources, allowing students to easily access, view, and download notes relevant to their subjects.

- **Interactive Event Calendar:**

Students can mark important events, exam dates, and assignment deadlines in a dynamic, personalized calendar interface.

- **GPA and CGPA Calculator:**

Students can calculate their academic performance across semesters based on subject marks and credit points.

- **Entertainment and Gamified Learning Tools:**

Quiz games are introduced to create a stress-free, fun learning environment that reinforces knowledge in an interactive way.

- **Course Learning Platform with Certification:**

Students can enroll in courses, complete modules, and earn certificates, enhancing their academic portfolio.

3.4 ADVANTAGES OF PROPOSED SOLUTION

1. Enhanced Student Engagement

Interactive Features:

Smart Sphere Classroom integrates quiz games, live chat support, and real-time notifications, creating an exciting and lively learning environment.

Active Participation:

The AI chatbot and gamified elements encourage students to participate more actively rather than being passive learners.

Continuous Motivation:

Points, badges, and achievements from quiz games help maintain student enthusiasm and drive throughout the academic term.

2. Timely Support and Communication

Instant Notifications:

Daily automatic reminders about assignments, tests, and events ensure that students are always up-to-date.

24/7 Assistance:

The AI-powered chatbot provides instant responses to students' academic doubts at any time, reducing dependency on teachers for minor queries.

Reduced Missed Deadlines:

Scheduled alerts greatly minimize the chances of students forgetting assignments or tests, thereby improving overall course completion rates.

3. Efficient Academic Management for Teachrs

Simplified Workflow:

Teachers can manage assignments, quizzes, student submissions, and classroom resources—all from one unified dashboard.

Centralized Resource Sharing:

Easy upload and sharing of lecture notes, assignments, and study materials ensure smooth academic operations.

4. Real-Time Feedback and Performance Analytics

Immediate Test Results:

Students no longer have to wait days for results—performance feedback is available immediately after test submission.

Analytics-Driven Insights:

Performance trends, GPA/CGPA calculations, and score analytics help both students and teachers identify strengths, weaknesses, and areas of improvement quickly.

5. Personalized Learning Support

Customized Academic Guidance:

The AI chatbot suggests resources, clears doubts, and helps students find targeted material based on their needs.

Self-Paced Learning:

Students can study and solve doubts at their own pace without being restricted to class timings.

6. **Cost-Effective and Accessible**

No Expensive Licensing:

By utilizing open-source tools and frameworks, the platform minimizes costs, making it ideal for institutions with limited budgets.

Accessible Anytime, Anywhere:

Students and teachers can access the platform from any device, supporting remote learning and hybrid educational models.

7. **Promotes Ethical and Fair Academic Practices**

Secure Data Management:

Implements secure authentication (JWT) and data privacy standards to protect user information.

Unbiased Evaluation:

Automatic test evaluation eliminates manual errors or biases in grading.

Fair Use of AI:

AI chatbot usage is ethical, with controlled and academic-focused responses only.

3.5 SYSTEM WORKFLOW

Smart Sphere Classroom is a secure, AI-integrated LMS designed to enhance online education for both students and faculty. Upon login, users access personalized dashboards based on their roles. Students can interact with the Gen AI chatbot for instant academic support, while faculty can resolve queries and manage learning tasks. Key modules include assignment submissions with automated Email.js reminders, gamified quizzes for interactive learning, and a test portal offering instant auto-evaluated results.

The platform also features a course module for enrollment and certification, managed by instructors. A custom calendar scheduler keeps users informed of upcoming deadlines and events. Additional tools like GPA/CGPA calculators, classroom folders, and real-time notifications ensure efficient communication, performance tracking, and a seamless learning experience.

3.5.1 WORKFLOW DIAGRAM FOR FACULTY ACCESS

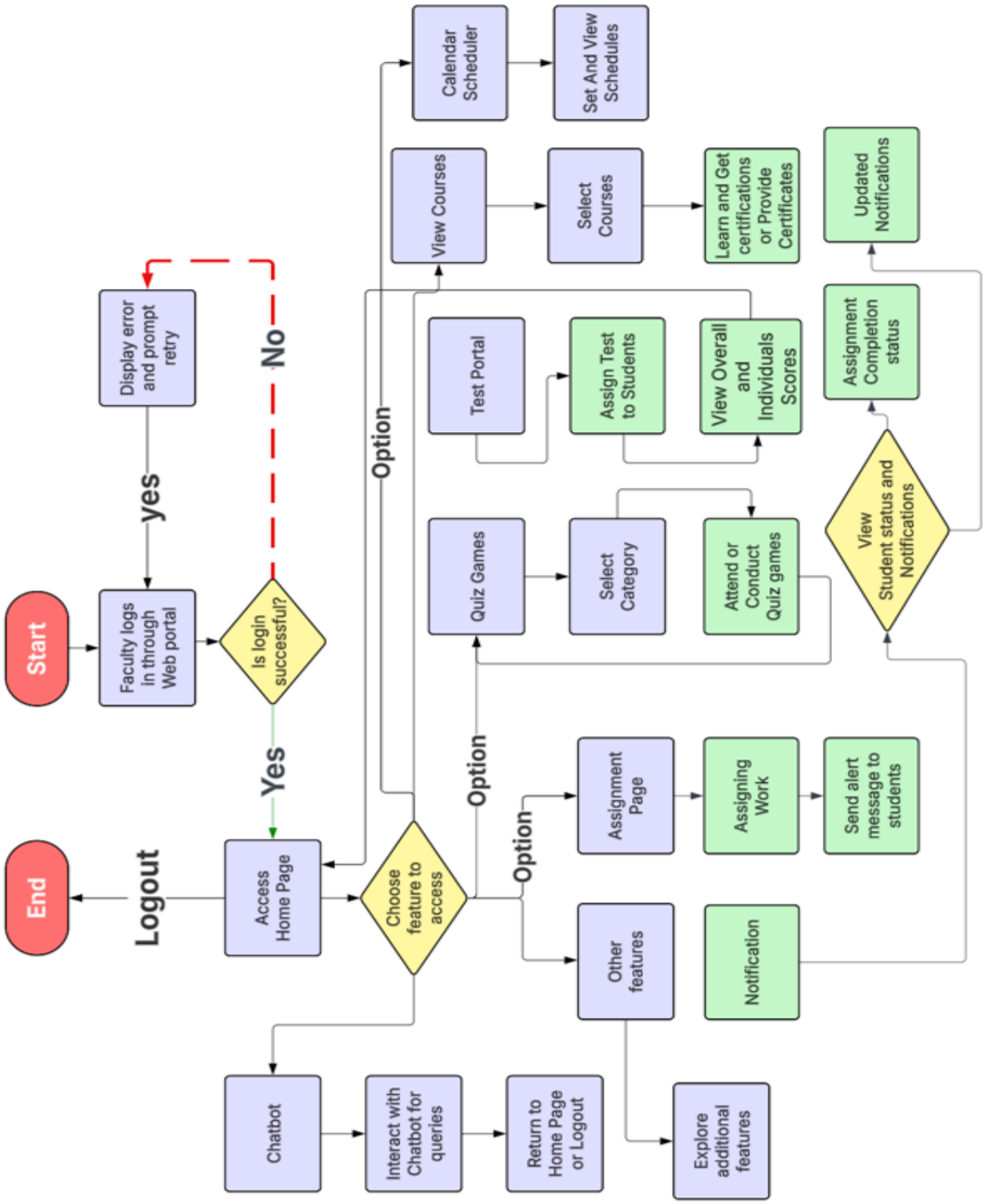

Fig.3.1 Workflow for Faculty Access

3.5.2 WORKFLOW DIAGRAM FOR STUDENT ACCESS

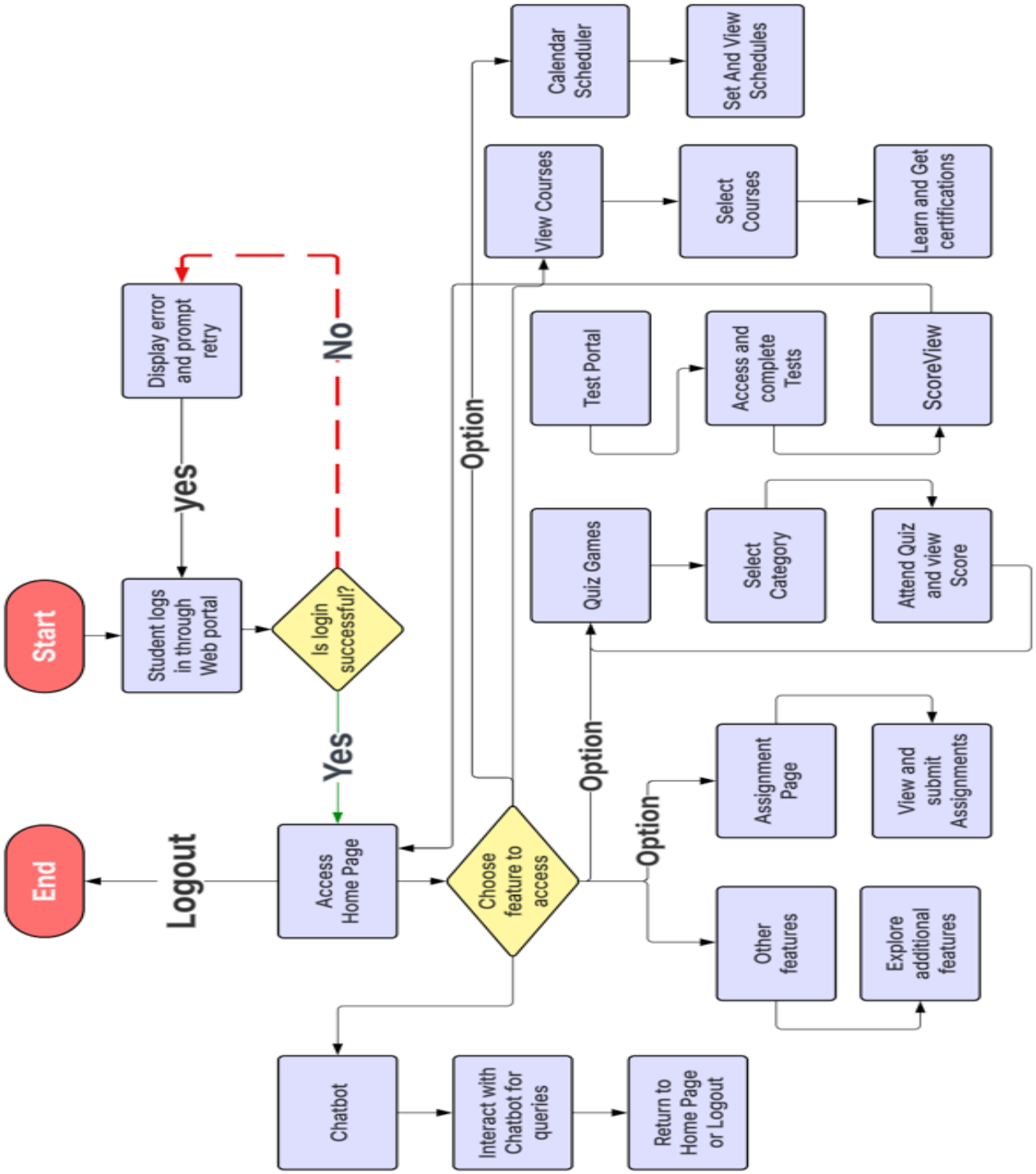

Fig 3.2 Workflow for Student Access

3.6 MODULES DESCRIPTION

3.6.1 CHATBOT MODULE (DOUBT CLARIFICATION SYSTEM)

The Chatbot Module offers instant academic support by answering subject-related queries using AI. Built with a Express.js and Gen Flash AI model, it processes student inputs via natural language understanding and returns accurate responses. Conversation history is stored securely in MongoDB for future analysis and improvement.

3.6.2 ASSIGNMENT MANAGEMENT MODULE

This module streamlines assignment creation and submission. Developed with Node.js and Express.js, it stores assignment data in MongoDB. Faculties can post assignments with deadlines, while students submit via a secure portal using Multer for file handling. The system tracks status and organizes tasks efficiently.

3.6.3 CLASSROOM FOLDER MODULE

Serving as a digital resource hub, this module lets faculties upload study materials, notes, and resources. Built with HTML, CSS, JavaScript, and Node.js, it uses Multer and MongoDB GridFS for file storage. Structured folders make materials easily accessible to students at all times.

3.6.4 CUSTOM NOTIFICATION MODULE

Powered by Email.js, this module sends personalized reminders about tasks and deadlines. Students can set notification intervals (e.g., every two hours). Preferences are stored in MongoDB to ensure timely and user-specific alerts.

3.6.5 TEST AND QUIZ PLATFORM MODULE

This module enables faculty to create online quizzes and tests. Developed using Node.js and Express.js with MongoDB, it supports multiple-choice quizzes, automatic evaluation, instant scoring, and leaderboard generation—promoting a competitive and transparent learning environment.

3.6.6 QUIZ GAME MODULE

This gamified module enhances learning with timed quizzes built using vanilla JavaScript and CSS animations. Students can test their knowledge across subjects, receive immediate scores, and track performance, blending education with engagement.

3.6.7 GPA & CGPA CALCULATOR MODULE

A frontend JavaScript tool for calculating semester GPA and cumulative CGPA based on subject grades and credit values. It helps students track and improve their academic performance with ease.

3.6.8 COURSE LEARNING PLATFORM MODULE

This module allows students to enroll in structured mini-courses and earn digital certificates upon completion. It supports progress tracking and encourages self-paced learning, expanding academic engagement beyond regular curriculum.

CHAPTER 4

SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE

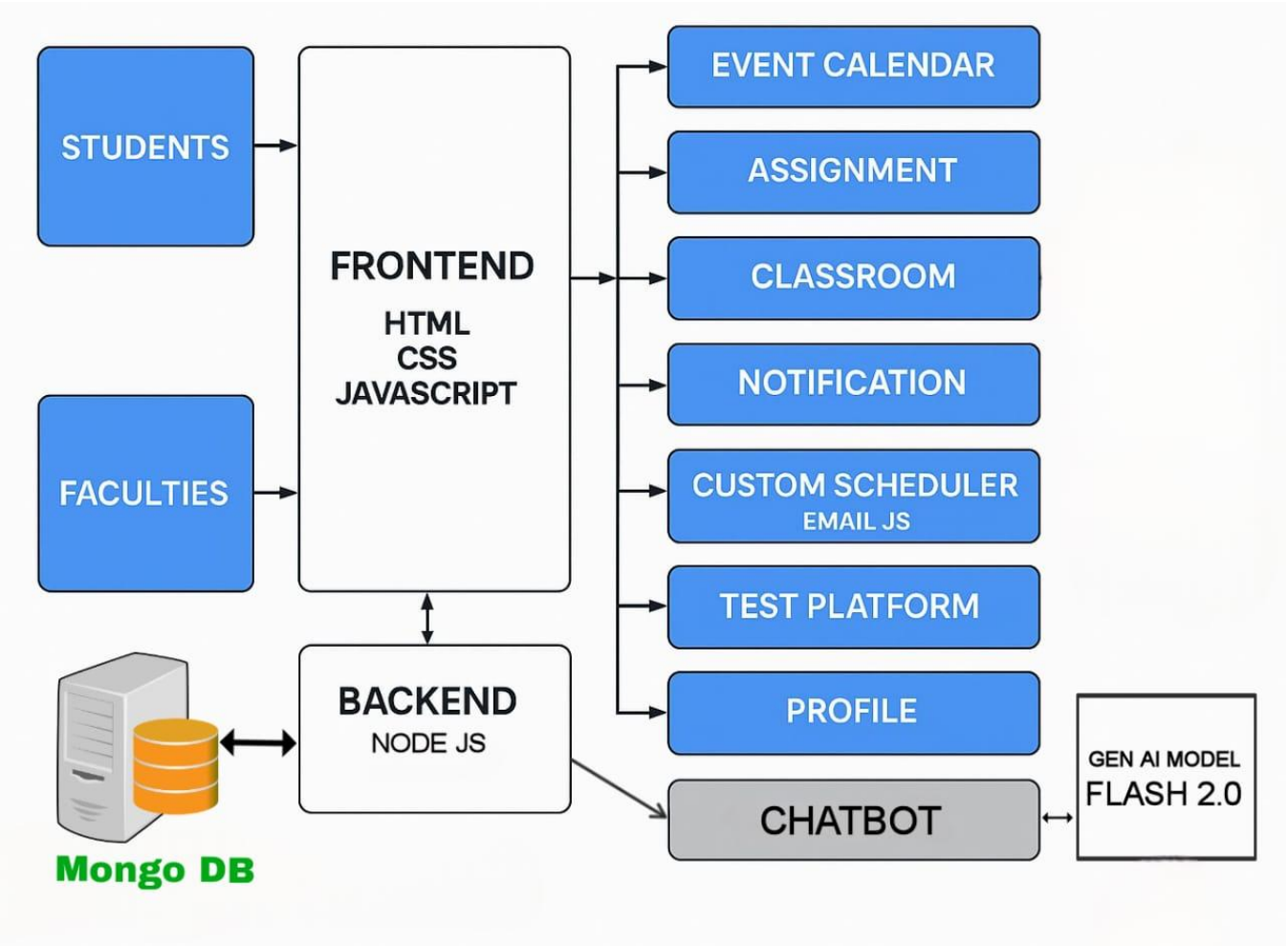

Fig 4.1 System Architecture

The architecture of the Smart Sphere Classroom is designed with a modular, scalable, and user-friendly approach to cater to the needs of both staff and students. The system follows a three-tier architecture comprising of Client Layer, Server Layer and Data Storage Layer.

4.1.1 CLIENT LAYER (FRONTEND):

The Client Layer of Smart Sphere Classroom serves as the direct interface between users (students and staff) and the system. It is responsible for collecting user inputs, displaying information, and interacting with the server through API calls. It ensures an intuitive, smooth, and responsive experience.

PROCESS:

1) USER AUTHENTICATION (LOGIN/SIGNUP):

- When the user opens the app, the first page presented is the Login or Signup Page.
- Users input their credentials. Credentials are validated locally (for format) and sent to the server for authentication.
- If successful, the user is redirected to their personalized dashboard.

2) MENUS (STUDENT/STAFF VIEW):

- After login, based on their user role (Student or Staff), they are shown a dynamic dashboard.
- The menus provides quick access to all system features: Courses, Notes, GPA/CGPA Calculator, and AI Chatbot.

3) COURSE LISTING AND COURSE DETAILS:

- When users click on "Courses," the client fetches the list of available courses from the server.
- Each course can be opened to view course descriptions, uploaded notes, and associated resources.

4) NOTES UPLOAD AND DOWNLOAD INTERFACE:

- Staff users can upload study materials (notes, PDFs, documents).
- Students can download notes for their respective courses.

- Upload/download buttons interact with backend APIs and provide success/error messages visually on the screen.

5) GPA/CGPA CALCULATOR INTERFACE:

- Users can input their semester-wise grades or scores.
- The frontend collects input, validates it, and either:

a. Calculates directly (if logic is on frontend), or

b. Sends to server for calculation (if backend logic exists).

6) EMBEDDED AI CHATBOT:

- An AI-powered chatbot is embedded in the interface (in a floating icon or sidebar).
- Users can type questions regarding courses, notes, or GPA calculations.
- The client sends the question to the chatbot API and displays the AI-generated response in real time.

7) USER FEEDBACK AND INTERACTIVITY:

- Every action (upload, download, GPA calculation, chat) gives instant feedback through pop-up modals, animations, or notifications to improve user engagement.
- The frontend ensures minimal loading time and dynamic updates without reloading pages.

8) SECURITY AND SESSION HANDLING:

- Frontend stores session tokens securely (typically in memory or secure storage) to maintain the login session.
- Unauthorized access to pages is blocked by redirecting to login if the session expires.

4.1.2 SERVER LAYER (BACKEND):

The server manages the application logic, handles user requests, and processes data.

PROCESS:

1) RECEIVING CLIENT REQUEST:

1. The server listens for incoming HTTP requests from the client (browser/app).
2. Requests can be:

a. Login/Signup

b. Upload or Download Notes or View Courses

c. Ask AI Chatbot a question

d. GPA/CGPA Calculation

e. Test Platform activities

3. The server captures important information like:

a. Request type (GET, POST, PUT, DELETE)

b. User session token or credentials

2) AUTHENTICATING USER:

1. Server checks whether the user is properly logged in.
2. If using JWT Tokens or Session Cookies:

a. Token is decoded and verified.

b. User role (staff/student) is checked.

c. If authentication fails:Server returns an error response: **"Access Denied. Please login again."**

d. If successful: “**Access to Home**”

3) IDENTIFYING THE REQUESTED SERVICE:

1. After authentication, the server identifies what the user wants:

a. If the request is related to Course Page → Route to Course Management Controller.

b. If it is about Notes Upload/Download → Route to Notes Controller.

c. If user interacts with AI Chatbot → Forward query to Chatbot Service.

d. If user inputs grades for GPA/CGPA Calculator → Send to Calculation Service.

4) PROCESSING BUSINESS LOGIC:

1. The respective backend controller processes the business logic:

a. Validate the input data (e.g., check file format for upload).

b. Perform actions:

a. Save data to database

b. Retrieve courses or notes

c. Call AI API for chatbot responses

d. Calculate GPA/CGPA based on formulas

2. If needed, the backend talks to database or external APIs.

5) SENDING RESPONSE TO CLIENT:

1. After completing the backend process, the server:

a. Prepares the correct response (success, data, download link, chatbot answer, etc.).

b. Sends the response back to the frontend.

2. The client (browser) receives the result and displays it to the user.

4.1.3 DATA LAYER (DATABASE AND STORAGE):

The data layer of the Smart Sphere Classroom system is designed to securely manage and store all critical information including user profiles, course details, uploaded notes, and system-generated data. The system utilizes **MongoDB**, a flexible and scalable NoSQL database, to structure and manage the data efficiently. The primary

collections maintained within the database include **Users** (which store information about both staff and students), **Courses** (containing course details and enrollments), **Notes Metadata** (capturing information such as file paths, associated courses, and uploader credentials), and optionally, **Chatbot Query Logs** for tracking and analyzing chatbot interactions to improve user support services. Uploaded files such as assignments and study notes are stored either on a **local server storage** or a **cloud-based storage solution**, depending on deployment configuration and scalability requirements. This organized and secure data layer ensures that the Smart Sphere Classroom platform operates with high reliability, accessibility, and data integrity.

4.2 TECHNICAL STACK

4.2.1 FRONTEND (CLIENT-SIDE)

The frontend of Smart Sphere Classroom is built using HTML5 for structured content and semantic markup, with CSS ensuring responsive, mobile-friendly design. JavaScript adds dynamic behavior and interactivity, enabling reusable, component-based interfaces. The Fetch API handles HTTP communication with the backend. JWT (JSON Web Tokens) is used for secure login and session management.

4.2.2 BACKEND (SERVER-SIDE)

The backend is developed with Node.js and Express.js to manage RESTful API requests. Express handles routing, authentication, assignment management, file uploads, test evaluation, and chatbot integration, ensuring scalability and reliable performance across system services.

4.2.3 DATABASE LAYER

MongoDB, a flexible NoSQL database, is used for storing unstructured data such as user profiles, assignments, tests, notes, course modules, and chatbot logs. Efficient indexing and schema design ensure fast data access and real-time operations.

4.2.4 NOTIFICATION SYSTEM

Notifications and reminders are managed using Email.js, a client-side email service that sends customized alerts without a backend mail server. Students and faculty receive scheduled emails for assignments, tests, and other key activities, enhancing engagement and time management.

4.2.5 AI CHATBOT SYSTEM

Integrated directly into the frontend using JavaScript, the AI Chatbot leverages the Gen Flash AI 2.0 model for advanced Natural Language Processing. Student queries are processed in real-time, and responses are fetched from the AI model. Chat logs can be stored in MongoDB to support future improvements.

4.3 CHATBOT AI WORKING AND DESIGN

The AI chatbot serves as a virtual assistant, helping users with course-related queries, GPA calculations, and platform navigation. With Generative AI 2.0 (e.g., GPT-4), the chatbot gains enhanced context understanding and conversational capabilities, enabling it to handle a wide range of queries beyond simple rule-based responses.

UNDERSTANDING QUERIES:

NATURAL LANGUAGE PROCESSING is the core technology for processing the user’s language and ensuring the chatbot can understand and respond to the students or staff appropriately. Here’s how NLP plays a crucial role:

User Input Processing

When a user (student or staff) submits a query, the chatbot uses NLP to break down and interpret the input. For example:

Query: "What is the due date for my math assignment?"

→ **Tokenization**: The sentence is split into individual components like “What,” “is,” “the,” “due,” “date,” “for,” “my,” “math,” and “assignment.”

→ **Part-of-Speech Tagging**: Identifies that "math" is a subject, "assignment" is a noun, and "due date" is a specific query.

→ **Named Entity Recognition (NER)**: Detects that “math” is a subject and "assignment" refers to a specific type of academic work.

Intent Recognition and Entity Extraction

After tokenizing and tagging the input, the chatbot performs intent recognition and entity extraction to identify the user’s goal and gather relevant information.

→ **Intent**: The intent behind the query is to find the due date for an assignment.

→ **Entities**: The subject, “math,” is extracted, and the keyword “assignment” tells the chatbot that the user is inquiring about an academic task.

With this, the chatbot knows the user wants information about the due date of a math assignment and can provide an appropriate response.

RESPONSE GENERATION:

Once the chatbot has understood the user’s query, it needs to generate a response. There are different ways to handle this and it would depend on the level of complexity and intelligence you wish to integrate into the system.

Rule-Based Responses

“When is my next class?”, the chatbot uses pre-programmed responses that are triggered when certain keywords or intents are detected. For example:

Query: “When is my next class?”

→ The chatbot identifies the intent ("class schedule") and provides a pre-configured response: “Your next class is on Monday at 10 AM.”

This approach works well for structured data or frequently asked questions. However, it’s limited in terms of flexibility since it can only respond to pre-defined inputs.

AI-Based Responses (Dynamic Generation)

For more complex or personalized queries, the chatbot could use AI-based models to generate responses dynamically, especially when information needs to be fetched or processed. Here’s how:

Dynamic Query Example: “What are the grades for my physics course?”

→ The chatbot can fetch this data from the backend (database) by connecting to the student’s records and dynamically forming a response based on the specific grades and course data available.

PERSONALIZED AND ADAPTIVE LEARNING:

One of the key advantages of Generative AI 2.0 is its ability to offer personalized experiences. It can learn from user interactions and adapt its responses accordingly.

Personalized Recommendations

The chatbot can recommend study materials or practice exercises based on the student’s past behavior or academic performance. **Example**: "I see you’ve been struggling with differential equations in your Math course. Here are some additional resources that might help you master this topic."

Real-Time Learning

Generative AI 2.0 can analyze ongoing interactions and provide real-time feedback or help. For instance, if a student is asking a lot of questions about a particular topic (like "chemistry"), the chatbot might recommend a deeper dive into that subject. If a

user is asking a lot of questions about assignments, grades, or courses, the chatbot adapts its responses based on the student's educational journey and learning needs, providing more depth or suggesting areas that might need attention.

4.4 DATABASE STRUCTURE AND WORKING

The Smart Sphere Classroom platform utilizes a well-structured NoSQL database (MongoDB) to store and manage all essential data components. The system architecture follows a collection-based structure, where each collection represents a functional entity. These collections are interrelated, providing seamless data access and efficient performance across various modules.

4.4.1 STRUCTURE:

Users Collection

This core collection manages data for all users—students, faculty, and administrators. Each document includes a unique user ID, email, role type, profile details, and references to enrolled or managed courses. User roles define access: students can submit assignments and view materials, faculty can manage courses and assessments, while admins handle system-wide settings and user privileges.

Courses Collection

Stores data on each course, including course name, code, description, assigned instructors, and enrolled students. It also references related assignments, notes, and exams, creating a centralized hub for course-related content and activities.

Assignments Collection

Contains details of all assigned coursework, such as title, description, deadline, and total marks. Each entry links to student submissions, storing file URLs, timestamps, and grades to support streamlined assignment tracking and evaluation.

Notes Collection

Stores shared learning materials uploaded by faculty or students. Each entry includes the note title, description, file URL, uploader ID, and associated course, promoting resource sharing and collaborative learning**.**

GPA Records Collection

Tracks students' academic progress by storing GPA data per semester. Each record includes student ID, term, completed courses, grades, and overall GPA, providing insights into academic performance.

Exams Collection

Manages scheduled course exams with data such as course reference, date, duration, location (if offline), and type (e.g., mid-term or final). Used by faculty and admins to organize and oversee assessments.

Chats/Conversations Collection

Logs user interactions with the Gen Flash AI 2.0 chatbot. Each document includes user ID, query, AI response, and timestamp, enabling context-aware assistance and supporting continual chatbot enhancement.

4.4.2 WORKING:

- **DATA FLOW**

The Smart Sphere Classroom platform operates on a structured data flow, where user interactions are captured and processed through specific collections in the backend. During registration, user credentials and roles (student, staff, or admin) are stored in the **Users** collection, enabling secure authentication and role-based access. Whenstudents enroll in courses, the **Courses** collection is updated accordingly, with

references stored in their user profiles. Faculty members can create assignments that are stored in the **Assignments** collection, while student submissions—including grades, submission times, and file URLs—are appended to the corresponding assignment records.Study materials such as lecture notes are uploaded by instructors or students and are stored in the **Notes** collection, with each file linked to its respective course. Exam schedules are managed by staff or administrators and stored in the **Exams** collection, including details like date, duration, and location, which students can access for preparation. Upon completion of academic terms, the platform calculates students' GPA based on completed courses and grades, storing the results in the **GPA Records** collection. Real-time interactions with the AI-powered chatbot are logged in the **Chats/Conversations** collection, preserving messages and responses for analytics and model improvement. This organized data flow ensures smooth operations, accurate tracking, and a responsive user experience across the entire platform.

CHAPTER 5

SOFTWARE DESCRIPTION

5.1 INTRODUCTION

The Smart Sphere Classroom is a feature-rich, web-based educational platform that integrates various academic functionalities to improve teaching, learning, and communication. The software system is designed with modular architecture and leverages modern web technologies to support students, faculty, and administrators in a unified environment. Below is the detailed breakdown of each major software component and its role in the platform:

5.2 USER AUTHENTICATION & ROLE MANAGEMENT

The system uses Node.js, JWT, and MongoDB for secure login and role-based access. Users are assigned roles—student, staff, or admin—which define their permissions. Students access courses and grades, while faculty manage content and track progress.

5.3 AI CHATBOT (GEN FLASH AI 2.0)

An AI chatbot built with JavaScript and Gen Flash AI 2.0 offers real-time academic support. It handles queries like deadlines and course info, enhancing student engagement through instant, natural-language responses.

5.4 COURSE MANAGEMENT

The Course Management module enables instructors to create, update, and manage course-related content, while students can enroll and track their academic journey. Implemented using **Node.js**, **Express.js**, and **MongoDB**, this module allows real-time updates of course progress, enrollment lists, and associated assignments or exams. **React** is used on the frontend to display course data interactively.

5.5 ASSIGNMENT AND NOTES UPLOAD/DOWNLOAD

This feature allows faculty to post assignments and upload study materials, while students can submit completed work and access notes. File handling is implemented using **Node.js**, and large files are stored via **MongoDB**. Notes and assignments are categorized per course and accessible through an intuitive **React.js** frontend interface.

5.6 GPA AND CGPA CALCULATOR

Students can calculate their semester-wise GPA and cumulative CGPA using this built-in tool. Its provide the Accurate result to user.

5.7 DASHBOARDS FOR STUDENTS AND FACULTY

Personalized dashboards are available for students and faculty. Students can view their enrolled courses, upcoming assignments, grades, and use the chatbot directly. Faculty dashboards allow for assignment and test management, student performance monitoring, and course editing. These dashboards are built using **React.js** and dynamically fetch user-specific data from **MongoDB** through secure backend APIs.

5.8 NOTIFICATION SYSTEM (EMAIL.JS)

To ensure timely updates, the platform employs **Email.js** to send real-time notifications about upcoming assignments, grade releases, and system announcements. Its supports scheduled alerts, triggered automatically based on deadlines or faculty actions. It ensures seamless academic communication and keeps students informed at all times.

CHAPTER 6

RESULTS AND DISCUSSIONS

6.1 OUTPUT AND DISCUSSIONS OF MODULES:

1) HOME PAGE:

Designed a centralized dashboard with navigation menus and real-time scheduled tasks.

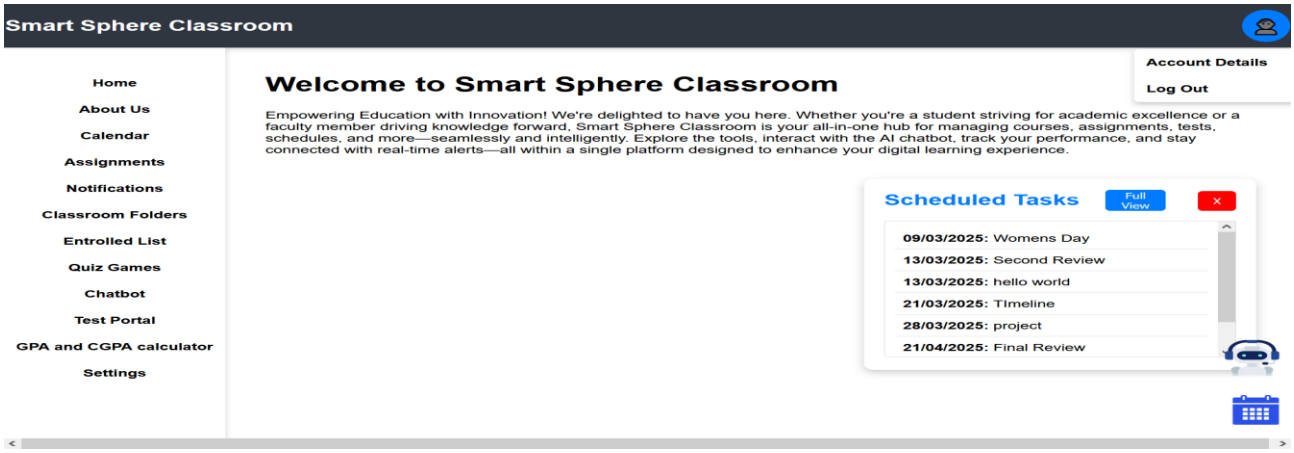

Fig.6.1 Home Page

This homepage offers users a seamless and intuitive experience, allowing them to navigate between different sections efficiently while keeping track of tasks and deadlines, thus enhancing productivity and engagement.

2) LOGIN AND SIGNUP PAGE:

Implemented secure login and signup forms with credential and JWT validation.

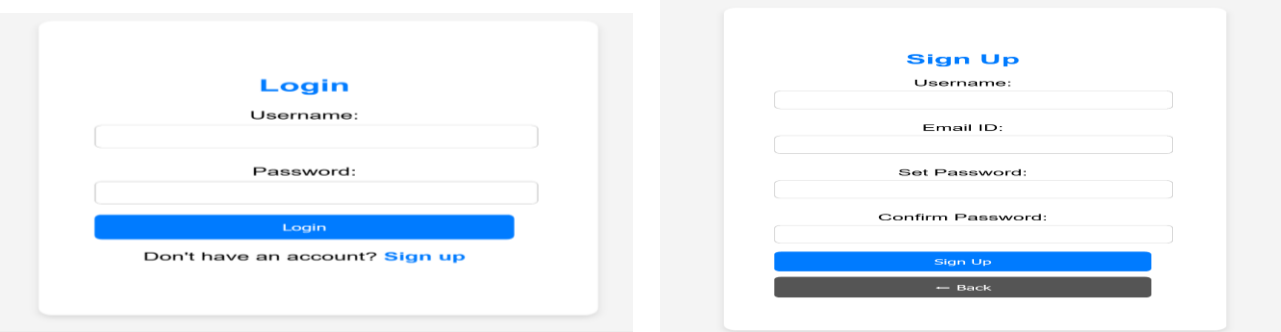

Fig 6.2 Login and Signup Page

This ensures a secure authentication process, enabling only registered users to access the platform while offering a smooth experience for new users to join.

3) SCHEDULING TASKS:

Built a task scheduling interface where users can set, save, and manage important dates.

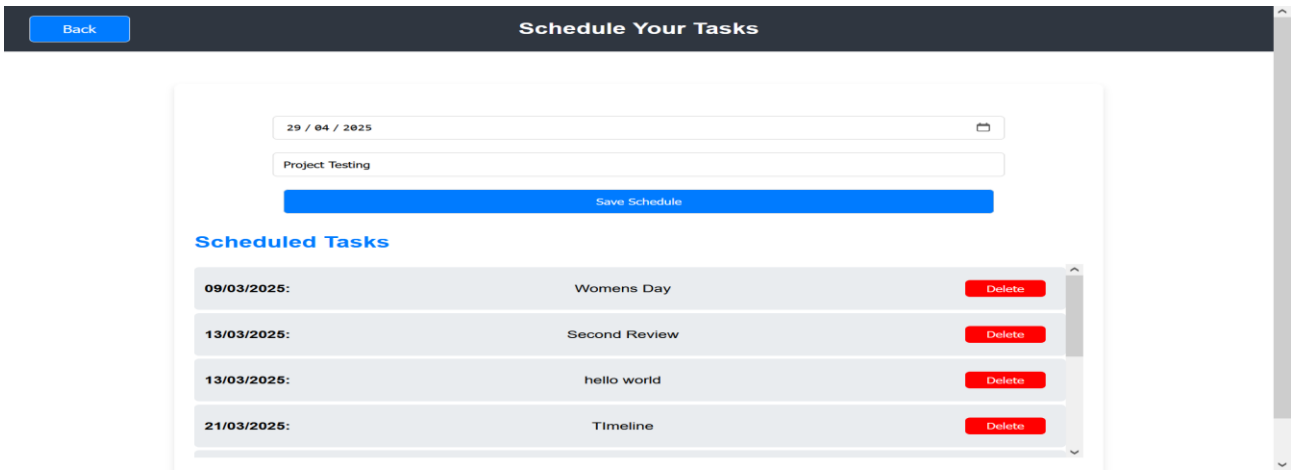

Fig 6.3 Scheduling Tasks

This helps users stay organized by planning assignments and activities in advance, reducing the chance of missed deadlines and improving time management.

4) ASSIGNED TASK PAGE FOR STUDENTS:

Created an assignment interface showing subjects, tasks, and file submission options.

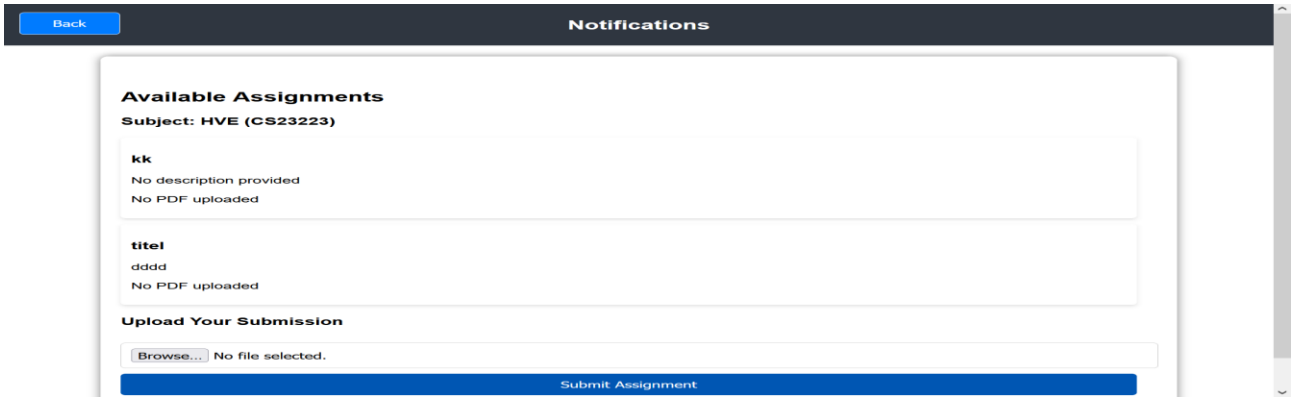

Fig 6.4 Assigned Tasks for Students

This creates an efficient workflow for students to manage and submit their academic work, and ensures clear communication of tasks from faculty.

5) COURSE ENROLLMENT:

Developed a module for browsing and enrolling in various online courses.

Fig 6.5 Course Enrollment

This module supports self-learning and skill development by giving students easy access to additional educational resources.

6) QUIZ GAMES:

Designed an interactive quiz system with multiple-choice questions and scoring.

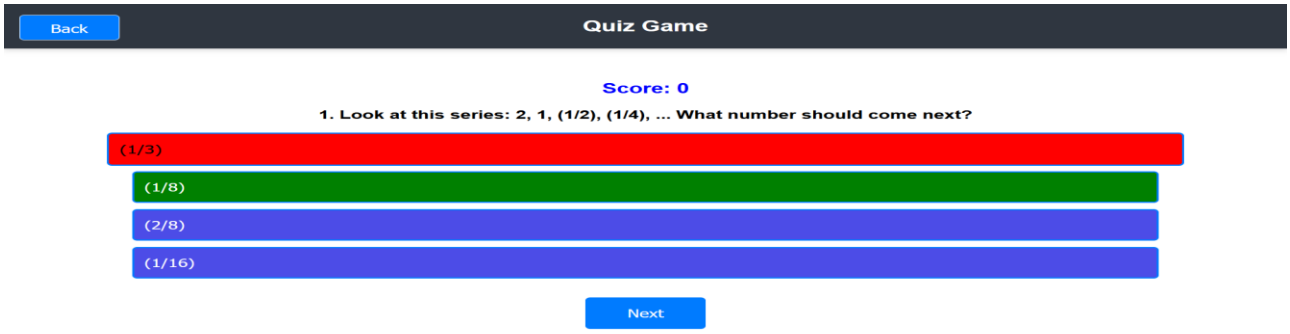

Fig 6.6 Quiz Games

This promotes active learning through a gamified approach, improving knowledge retention and encouraging regular practice in an engaging way.

7) SCHEDULED EXAMS:

Built an exam list interface showing upcoming exams, codes, and access links.

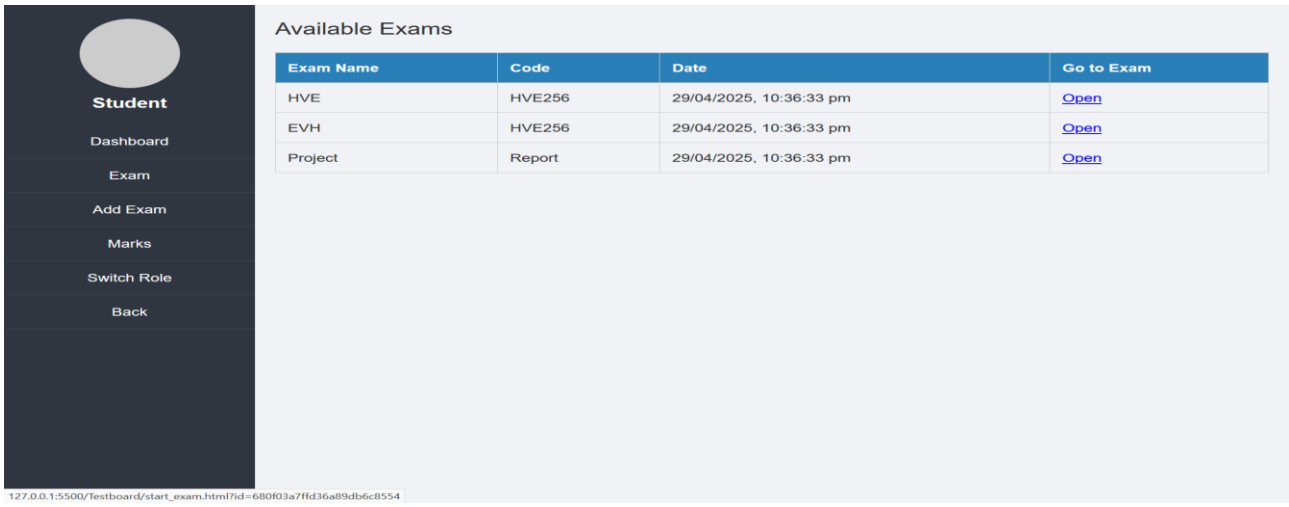

Fig 6.7 Scheduled Exams

This helps students plan and attend their exams on time while ensuring they are well-informed about their exam schedules and formats.

8) ASSIGNMENT ASSIGNING PORTAL FOR FACULTY:

Created a faculty panel to assign new tasks, upload files, and manage submissions.

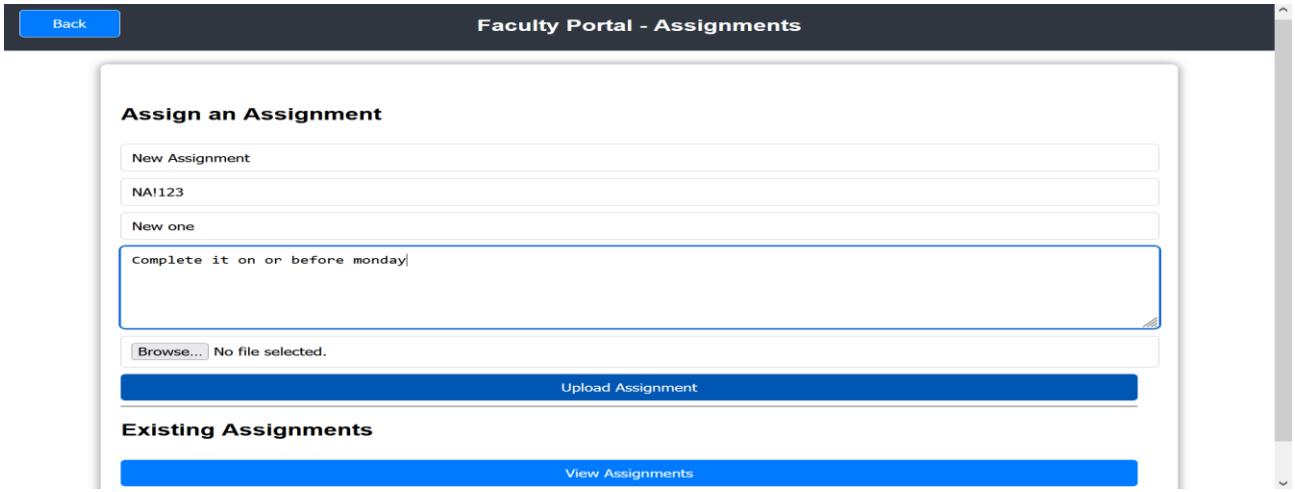

Fig 6.8 Assignment Assigning portal for Faculty

This simplifies the process for teachers to assign coursework and ensures a consistent, organized way to communicate academic tasks with students.

9) GPA AND CGPA CALCULATOR:

Built a calculator to enter subject grades and compute GPA/CGPA.

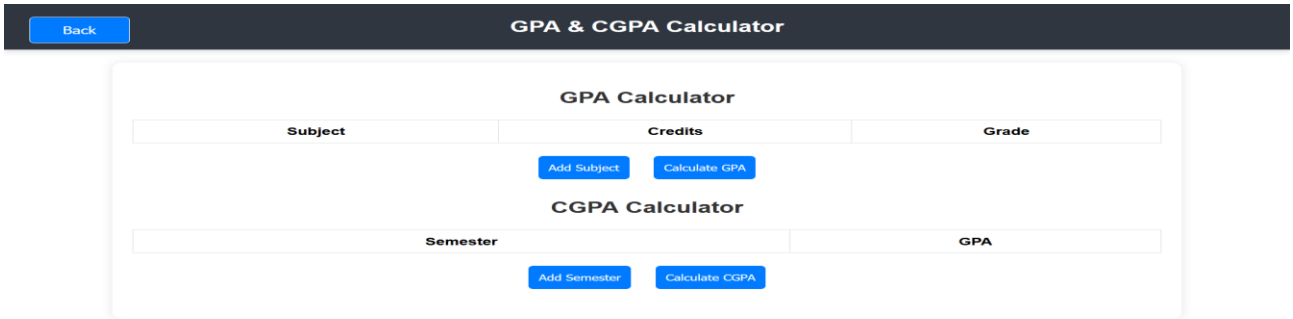

Fig 6.9 GPA and CGPA Calculator

This tool allows students to assess their academic standing easily, enabling them to monitor performance and take steps to improve if necessary.

10) ABOUT PAGE:

Designed an About page to describe the platform, its mission, and main features like automation and AI tools.

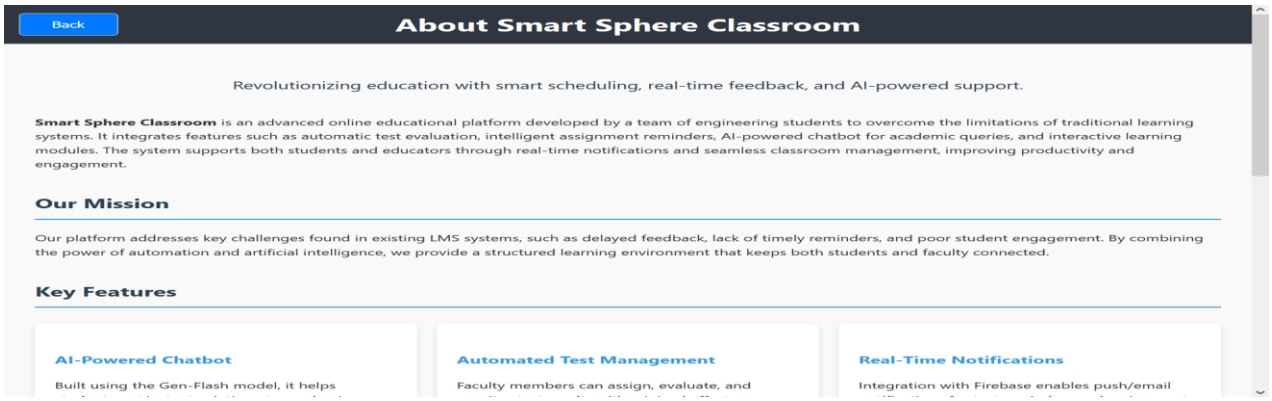

Fig 6.10 About Page

It introduces users to the platform’s goals and builds trust. The overview improves clarity on what the system offers and why it’s built.

11) CHATBOT:

Implemented a chatbot assistant to help students with queries related to subjects, exams, and platform usage.

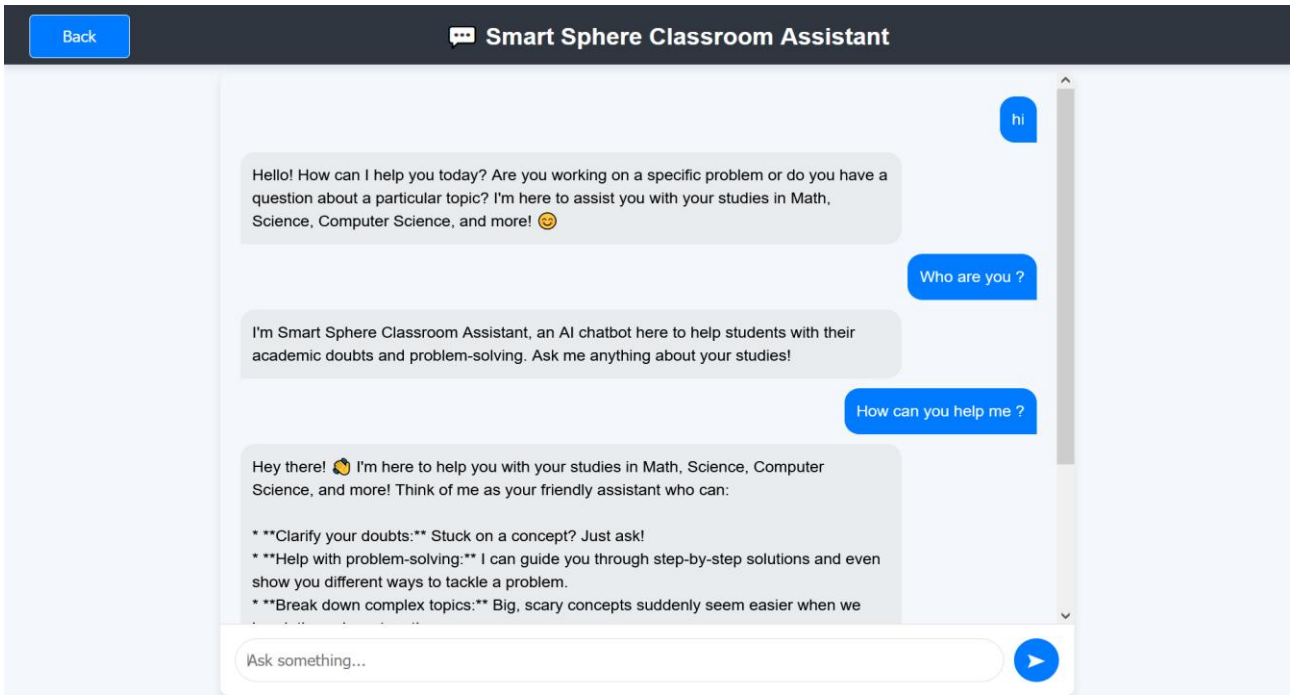

Fig 6.11 Chatbot

This reduces dependency on faculty by offering instant support. It enhances the user experience with quick, intelligent responses.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 PROJECT SUMMARY

The project titled Smart Sphere Classroom: An Intelligent and Integrated Educational Platform was initiated with the objective of enhancing the online learning experience by integrating modern technologies such as AI, automation, and real-time communication into a unified solution. Unlike traditional Learning Management Systems (LMS) like Moodle, Blackboard, and Google Classroom, Smart Sphere Classroom addresses key limitations by introducing AI-powered doubt clarification through a Gen-Flash-based chatbot, automated test evaluation with instant grading, and custom scheduling features with real-time alerts via Email.js. The platform offers a centralized interface for managing assignments, tests, schedules, course materials, and student engagement tools like gamified quizzes. Additionally, built-in GPA and CGPA calculators provide students with simplified academic tracking. The successful implementation of all planned features not only fulfilled but exceeded the initial project goals, resulting in a highly functional platform that holds strong potential for real-world deployment in educational institutions to support paperless learning and improve academic communication.

7.2 FUTURE WORK

Our initiatives for the future include giving the platform mobile compatibility, integrating with outside learning resources, and extending the capabilities of the chatbot. Long-term, we want to look into using student data analytics to customize education experiences and ensure that our methods are both ethical and sustainable.

APPENDIX – A

HOME PAGE :

Upon successful completion of the login or signup process—secured through **JWT (JSON Web Token)** authentication—the user is redirected to the **Home Page** of the *Smart Sphere Classroom* platform. This central dashboard serves as the primary navigation hub for all users, displaying personalized content based on their roles (student, faculty, or admin).

CODE :

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Smart Sphere Classroom</title>
    <link rel="stylesheet" href="style.css">
    <script defer src="script.js"></script>

    <style>
        .calendar-icon {
            position: absolute;
            bottom: 20px;
            right: 10px;
            font-size: 50px;
            color: #007bff;
            cursor: pointer;
            transition: transform 0.3s ease-in-out;
        }
```

```
.calendar-icon:hover {
    transform: scale(1.2);
}
#chatbot-icon{
    position: absolute;
    width: 60px;
    height: 60px;
    bottom: 100px;
    right: 9px;
    transition: transform 0.5s ease-in-out;
}
#chatbot-icon:hover{
    transform: scale(1.2);
}
/* Popup Styling */
.popup {
    position: absolute;
    top: 50%;
    /* Start below */
    left: calc(50% + 50px);
    /* Start beside the icon */
    transform: translate(-50%, -50%) scale(0);
    width: 350px;
    background: white;
    padding: 20px;
    border-radius: 10px;
    box-shadow: 0 4px 10px rgba(0, 0, 0, 0.2);
    display: none;
```

```
    opacity: 0;
    transition: transform 0.9s ease-in-out, opacity 0.9s ease-in-out;
}
.popup.show {
    display: block;
    transform: translate(-105%, -70%) scale(1);
    /* Move above the icon */
    opacity: 1;
}
.popup h2 {
    margin-top: 0;
    color: #007bff;
}
.close-btn {
    background-color: red;
    color: white;
    border: none;
    padding: 8px 15px;
    border-radius: 5px;
    cursor: pointer;
    float: right;
    transition: transform 0.5s ease-in-out;
}
.close-btn:hover {
    transform: scale(1.10);
}
.schedule-list {
    max-height: 200px;
```

```
    overflow-y: auto;
    text-align: left;
    padding: 10px;
    border: 1px solid #ddd;
    border-radius: 5px;
}
.schedule-list::-webkit-scrollbar {
    width: 6px;
}

.schedule-list::-webkit-scrollbar-thumb {
    background: #007bff;
    border-radius: 10px;
}
.schedule-item {
    padding: 8px;
    border-bottom: 1px solid #ddd;
    animation: slideIn 0.3s ease-in-out;
}
@keyframes slideIn {
    from {
        transform: translateX(-20px);
        opacity: 0;
    }
    to {
        transform: translateX(0);
        opacity: 1;
    }
```

```
        }
        #schedulefullviewbtn {
            position: absolute;
            top: 18px;
            border-radius: 5px;
            border: none;
            right: 90px;
            padding: px;
            background-color: #007bff;
            color: aliceblue;
            width: 60px;
            transition: transform 0.5s ease-in-out;
        }
        #schedulefullviewbtn:hover {
            transform: scale(1.10);
        }
    </style>
</head>
<body>
    <nav class="navbar">
        <div class="logo"><h1>Smart Sphere Classroom</h1></div>
        <!-- <div class="user-icon">&#128100;</div> -->
        <div class="Acc-circle"></div>
        <div class="user-menu">
            <div class="user-icon" onclick="toggleDropdown()"><img
src="userimg.png"></div>
            <div class="dropdown-menu" id="dropdown-menu">
                <a href="#">Account Details</a>
```

```
            <a href="#" onclick="logout()">Log Out</a>
        </div>
    </div>
</nav>
<div class="sidebar">
    <ul>
        <li><a href="index.html">Home</a></li>
        <li><a href="Aboutus/aboutapp2.html">About Us</a></li>
        <li><a href="calendar/calendar.html">Calendar</a></li>
        <li><a href="Assignment/Faculty.html">Assignments</a></li>
        <li><a href="Assignment/student.html">Notifications</a></li>
        <li><a href="Classroom/clsroom.html">Classroom Folders</a></li>
        <li><a href="Courseplatform.html">Entrolled List</a></li>
        <li><a href="QuizGames/quizpage.html">Quiz Games</a></li>
        <li><a href="Chatbot/chatbot.html">Chatbot</a></li>
        <li><a href="TestDashboard.html">Test Portal</a></li>
        <li><a href="Calculator/resultcalci.html">GPA and CGPA calculator</a></li>
    </ul>
</div>
<div class="content">
    <h1>Welcome to Smart Sphere Classroom</h1>
    <p>Select an option from the menu to get started.</p>
</div>
<div class="popup" id="schedulePopup">
    <button class="close-btn" onclick="closePopup()">X</button>
    <h2>Scheduled Tasks</h2> <button id="schedulefullviewbtn" onclick="movetoscheduler()">Full View</button>
```

```
        <div class="schedule-list" id="taskList"></div>
    </div>
    <script>
        function fetchSchedules() {
            fetch("http://localhost:2000/events")
                .then(response => response.json())
                .then(data => {
                    let popup = document.getElementById("schedulePopup");
                    let taskList = document.getElementById("taskList");
                    taskList.innerHTML = "";
                    if (data.length === 0) {
                        taskList.innerHTML = "<p>No upcoming schedules.</p>";
                    } else {
                        data.forEach(event => {
                            let eventDate = new Date(event.date);
            let formattedDate = eventDate.getUTCDate().toString().padStart(2, '0') + '/' +
                                (eventDate.getUTCMonth() + 1).toString().padStart(2, '0') + '/' +
                                eventDate.getUTCFullYear();
                            let item = document.createElement("div");
                            item.classList.add("schedule-item");
                            item.innerHTML = `<strong>${formattedDate}:</strong>
${event.desc || "No description"}`;
                            taskList.appendChild(item);
                        });
                    }
                    let icon = document.getElementById("calendarIcon");
                    let iconRect = icon.getBoundingClientRect();
                    popup.style.top = `${iconRect.top - 20}px`;
```

```
            popup.style.left = `${iconRect.right + 10}px`;
            popup.classList.add("show");
        })
        .catch(error => console.error(" Error fetching events:", error));
    }
    function closePopup() {
        let popup = document.getElementById("schedulePopup");
        popup.classList.remove("show");
    }
    function movetoscheduler() {
        window.location.href = "calendar/calendar.html";
    }
    function movetochatbot() {
        window.location.href = "Chatbot/chatbot.html";
    }
</script>
<img src="img/robot-assistant.png" alt="Chatbot" id="chatbot-icon" onclick="movetochatbot()">
<img src="img/icons8-calendar-50.png" alt="Calendar" class="fas fa-calendar-alt calendar-icon" id="calendarIcon"
    onclick="fetchSchedules()">
</body>
</html>
```

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